

NAME (PLEDGED):_____ POINTS: ____ / 15

1. What is the main focus of procedural programming?
2. What is the main focus of object-oriented programming (OOP)?
3. What are three key principles of OOP? Can you think of an example for each?
4. In the light-switch example, how does the `Switch` communicate with the `Light`?
5. What is encapsulation in OOP?
6. What is a class in C++?
7. What is an object in C++?
8. How does the car example illustrate interface vs. implementation?
9. Why might OOP be unnecessary for simple programs?
10. Is this definition allowed? Will it compile?

```
class foo {  
    // nothing to see here  
};
```
11. What does the `const` keyword after the function name mean?

```
double getBalance() const  
{ return balance; }
```
12. If `Car` is a class with a member function `Car(int cylinders)`, how could you construct a `Car` object `chevy` with 4 cylinders in the `main` function? Write the statement.

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13. You cannot get the `Car` object data given the following definition:

```
class Car{  
    private:  
        int cylinders;  
    public:  
};
```

Write a new member function `getCylinders` for the class that allows you to retrieve the number of cylinders.

14. Write a statement to print the number of cylinders of a `Car` object with 4 cylinders called `chevy`.

15. How many `Car` objects can you create in the same `main` program?